

Yusuf Gadelrab

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Authorized to work in the U.S. for any employer without sponsorship (U.S. Lawful Permanent Resident).

EDUCATION

San Jose State University | San Jose, CA

B.S. Computer Science, Applied Mathematics track | Expected May 2028

- Quantitative coursework: Probability & Statistics, Linear Algebra, Discrete Mathematics, Data Structures & Algorithms.
- Lynbrook High School — Valedictorian, 4.0 GPA (2024).

QUANTITATIVE EXPERIENCE

IBM SkillsBuild — AI Experiential Learning Lab | Remote

Quantitative NLP Project Lead | Jan - May 2026

- Built an NLP equity-scoring model on **IBM Watson** that converts financial news into directional signals, reaching **78% directional accuracy** on a six-month backtest.
- Automated ingestion of **50+ live market sources daily** into a feature pipeline scoring equities across five quantitative dimensions.
- Cut analyst research time **60%** by automating source aggregation, extraction, and signal generation end to end.

HwyHaul | San Jose, CA

AI / Agent Engineering Intern | May - July 2026

- Automated invoice and load-document intake for a logistics operator, converting unstructured carrier messages into structured billing records and **cutting entry time from ~90s to under 2s per load**.
- Built the extraction service in **FastAPI** with a deterministic fallback path, a full test suite, and versioned schema migrations for pilot deployment.

SJSU CSEd Research Lab (Dr. Ethel Tshukudu) | San Jose, CA

Undergraduate Researcher | Aug 2024 - Present

- Ran an **IRB-approved mixed-methods study of 60 participants** — designed the pre/post instruments and ran paired-sample t-tests, Wilcoxon signed-rank, and independent-sample t-tests to establish **statistically significant pre-to-post gains** in programming confidence, computing identity, enjoyment, and motivation, with novices gaining significantly more in confidence than experienced programmers.
- Co-authored a **peer-reviewed paper accepted to ACM SIGCSE TS 2026** (doi:10.1145/3770761.3777339).

Silicon Valley Entrepreneurship Club | San Jose, CA

Technical Ops & Web Lead | Dec 2025 - Present

- Secured **4 corporate sponsors** and expanded the operating budget **35%**, owning the sponsor pipeline from prospecting through negotiated terms.
- Manage event budgeting and reporting; built and maintain the club's full-stack site (sventclub.org), raising engagement **40%**.

Kappa Sigma, SJSU | San Jose, CA

Board Treasurer | Aug 2025 - Present

- Own the budget, dues collection, and financial reporting for a **50+ member** organization across **10+ events** per term.

QUANTITATIVE PROJECTS

Systematic swing-trading research | Python, Pandas, walk-forward validation, block bootstrap, Alpaca API

- Built a screening and backtest engine, then **walk-forward validated** candidate setups instead of reporting in-sample results: an anchored-VWAP reclaim setup held at **+0.117R over 4,933 trades** on a **129-symbol, 11-sector, 10-year universe** (block-bootstrap 95% CI [+0.057, +0.174]).
- Benchmarked the signal against a null model rather than against zero**: a risk-matched systematic entry already captures +0.086R of the same long drift, so I reported the signal's own incremental contribution as **+0.030R with a CI crossing zero** — and graded the strategy MARGINAL rather than tradeable.
- Rejected more hypotheses than I accepted, on the evidence**: an opening-range-breakout proxy came out **negative (-0.28R over 412 trades)** and a volatility-contraction breakout came out **breakeven-to-negative (-0.09R)**; regime (SPY > 200SMA) and price-location filters were both tested and dropped after they *reduced* expectancy on the same data.
- Wrote the promotion gate into code — the engine **refuses a live broker connection** until a paper track record clears a 60-trade, positive-expectancy threshold, so the risk decision cannot be made by impulse.

FreightDesk — back-office automation for freight brokers | Python (zero-dependency, stdlib HTTP services) — 66 passing tests

- Modeled unit economics against a **\$1,800–2,200/mo** in-house AR-clerk cost baseline to derive defensible pricing.
- Built three composable services (carrier intake, dispatch triage, AR billing) and a **428-record FMCSA-verified** broker dataset for market sizing.

DIRA — static security scanner | Python, OSV.dev, CycloneDX/SPDX — 138 passing tests

- Audited all **169 findings** the scanner produced across four repositories, narrowed one rule that was wrong on **17 of 17** hits, and re-ran to **84 findings each traced to a verified underlying fact**.

PUBLICATIONS

- Tshukudu, E., Shah, N., Kieu, T., Deeb, L., Venkateswaran, H., Ghai, A., **Gadelrab, Y.**, & Hada, P. (2026). "Exploring Bilingual Coding for Inclusive Computer Science Learning." *SIGCSE TS 2026*, ACM, 2 pages. doi:10.1145/3770761.3777339

SKILLS

- Quantitative**: Python (Pandas, NumPy, VectorBT), SQL, backtesting and walk-forward validation, block-bootstrap confidence intervals, expectancy and R-multiple analysis, position sizing, risk modeling, NLP sentiment analysis.
- Engineering**: FastAPI, React, Next.js, Git, Docker, pytest, IBM Watson Studio, local LLM orchestration.
- Markets**: Equities, ETFs, options, futures, FX; technical and macro analysis.
- Spoken**: Arabic (native), English (native).